INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

STERILIZER, SURGICAL INSTRUMENT AND DRESSING

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers two types of table top sterilizers suitable for sterilizing dental and surgical instruments and dressings.

Salient Characteristics:

The unit shall be a table top sterilizer, electrically heated, automatically controlled, high speed, pressure, single shell, encased in a cabinet, suitable for sterilizing unwrapped, wrapped packed, and liquid products in hospital, laboratory, scientific and medical office environments.

The sterilizer shall consist of a cabinet, sterilizing chamber, water reservoir, electric heater, control, and valves, and shall have at least one easily removable tray and rack fabricated of corrosion-resisting metal.

Maximum overall dimensions and minimum chamber dimensions of the sterilizer shall be as follows:

	Type I	Type II
Overall length:	24 inches	24 inches
Overall width:	20 inches	20 inches
Overall height:	18 inches	18 inches
Chamber diameter:	7-7/8 inches	9-7/8 inches
Chamber usable depth:	14 inches	15-1/4 inches

AMSC/NA FSC 6530

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The sterilizer shall have the following operating controls, instruments, and visible signal devices mounted in the face panel for ease of reading and accessibility:

* A temperature controller/indicator which shall control and indicate the temperature between 212-270°F in the chamber, and shall start the timing phase of the cycle at set temperature.

* A cycle timer which shall control the period of time that the load will be exposed to steam at set temperature.

* A processing or condition light which shall indicate the chamber is being heated to operating temperature. It shall come on when the door is in the locked position and time is set on the cycle timer.

* A sterilize light which shall come on during a pressurized cycle when temperature has been attained.

* A light indicating a drying phase which shall come on when the door is opened and time is set on cycle timer for drying a load at low temperature.

* A reset button which shall restore operation after overheat switch has tripped to prevent overheating when water is low in chamber. As an alternate, the unit shall have a suitable fail-safe method to render the unit inoperable until the overheat condition has been eliminated.

The sterilizer shall include a recorder which provides a permanent paper tape record of each cycle or the critical exposure time. Shall have a one degree resolution. The temperature digital readout and analog representation shall be in degrees Fahrenheit or Centigrade (selectable). The unit shall print, "cycle start", "date", exposure time, exposure temperature, cycle number and actual exposure time at temperature.

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Cycle selection shall be for at least wrapped goods, unwrapped instruments and liquids. Cycle phase timers shall be a minimum of 0-30 minutes for drying phase and 3-90 minutes for exposure phase, operator selectable with factory preset default settings as follows:

a. Instrument or soft goods (unwrapped instrument (flash) cycle): 3 minutes exposure at 275°F and 15 minutes drying time.

b. Liquid cycle: 30 minutes exposure at 250°F with no drying time. The unit shall automatically control the return of the chamber to atmospheric pressure at a rate of approximately 4.5°F per minute to prevent liquid boilover.

c. Wrapped goods cycle: 10 minutes exposure at 250°F, with adjustable dry time.

There shall be an audible signal for cycle complete. A visible indication of an overheat condition shall be displayed to notify the operator.

The sterilizer shall be fabricated of corrosion-resisting metal with the exception of minor parts such as knobs, handles, controls, insulation, etc. As an alternate, the outer cabinet may be constructed of a combination of high temperature polysulfone, heat resistant ABS or a suitable material, finished to resist corrosion.

The unit shall be capable of completing at least ten sterilization cycles, with load, on one filling of the reservoir. It shall further be designed to return the residual steam and condensate in the chamber to the reservoir following a sterilization cycle. The reservoir shall be completely enclosed within the cabinet. The fill opening for the reservoir shall be located on the top of the cabinet. The reservoir shall have a cover to adequately prevent inadvertent spillage of distilled water during pitch and roll movements when installed aboard naval vessels or other transport modes. A drain connection for emptying the reservoir shall be accessible from the exterior of the cabinet.

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The operating temperature range shall be 250°F to 270°F. The sterilizer shall be capable of attaining and maintaining a temperature of 270°F for five minutes, and 250°F for 20 minutes. The sterilizer control system shall control chamber temperature to within +6°F to -2°F of the set point.

The unit shall be constructed to satisfactorily withstand a chamber pressure of 60 psi and shall include a safety valve set at a pressure between 38 psi and 45 psi.

The sterilizer shall have a chamber door mechanism that will offer resistance to the user such that the door cannot be opened until chamber pressure is vented and reduced to at least 3 psig. As an alternate, it shall be necessary to manually activate the chamber exhaust vent, before the door latch can be released.

A three conductor line cord at least six feet in length, equipped with a three prong male plug shall be furnished.

Average cycle time with 2 pounds of unwrapped instruments shall not exceed 28 minutes from a cold start and 20 minutes from a hot start.

The sterilizer shall comply with the ASME Pressure Vessel Code, and UL Standard 544 and shall be listed by ETL, UL or other nationally recognized testing laboratories.

The unit shall be designed and constructed to meet the following regulatory codes:

a. C.S.A. Certification and Listing
b. I.E.C. 601.1 Compliance on 220 VAC
c. Closure shall be in compliance with U.G.35 A.S.M.E. B & P code, Section VIII, Div. 1.

The maximum load size which the unit shall be capable of processing shall be:

a. Eight lbs. unwrapped instruments
b. Four lbs. double wrapped instruments
c. Three 500 ML beakers 75 percent full
d. Linen pack which contains two towels, four sponges, one tape sponge and one cover

The unit shall be designed to operate in altitude from sea level to 10,000 feet.

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Electrical requirements. The sterilizer shall be capable of operating on 115/230 volts, 50/60 hertz.

Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

Operational and Service Data Manuals shall be furnished as follows:

a. Information content:

The contractor shall furnish manuals, handbooks, and/or brochures containing complete operation, installation, and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language, and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name, and quantity required. All service notes, service memos, etc. pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, those furnished to service engineers (or servicemen) who normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

b. Quantities, distributions and schedules:

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, only a total of two service manuals need to be supplied to each consignee. However, two operational manuals still need to be supplied with each unit.

Workmanship. The sterilizer shall be free from defects which detract from its appearance or impair its serviceability.

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Unit. Each (EA). One sterilizer, as specified, constitutes one unit. Each sterilizer shall be packaged in a suitable commercial container which shall provide adequate protection against corrosion and physical damage during shipment and handling. Suitable noncorrosive cushioning material shall be used, if necessary.

Storage characteristics: Item may be subjected to storage in an uncontrolled environment (temperature and humidity) for extended periods of time, e.g., 5 to 10 years. If the item can only be stored in a controlled environment, the contractor shall submit information with the proposal/bid which clearly explains the environmental limits within which the item shall be stored. The contractor shall provide adequate packaging, recommended repair parts (parts subject to deterioration during storage, e.g., rubber goods, tubing, batteries), and maintenance/inspection schedules to prevent problems in operation after extended periods of storage. For items requiring periodic inspections/maintenance while in storage, the contractor will provide a means to access the equipment/components without permanently destroying the packaging container, e.g., provide reusable shipping/storage containers. The cost of providing such packaging, if different from the usual method of shipping, will be separately identified in the proposal/bid. This will allow the Government to determine the cost effectiveness of this requirement versus the Government performing its own packaging for storage.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an Acceptable Quality Level (AQL) of 2.5 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one sterilizer.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one sterilizer.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE 1: The following National Stock Numbers are covered by this document.

Type I 6530-01-189-6960
Type II 6530-01-120-2658

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NOTE 2: Underwriters Laboratories (UL) standards are available from UL, Publications Stock, 333 Pfingsten Road, Northbrook, IL 60062.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - OSS
USPHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Project Number: 6530-2147

Location: STERMC1B.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54597A	2 DOCUMENT DATE (YYMMDD) 941216

3 DOCUMENT TITLE
STERILIZER, SURGICAL INSTRUMENT AND DRESSING

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	c TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MAP	b TELEPHONE (Include Area Code) (1) Commercial (215)737-8431 (2) AUTOVON 444-8431
c ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041 3466 Telephone (703) 756-2340 AUTOVON 289-2340

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